In Situ Growth of Tin Oxide Nanowires, Nanobelts, and Nanodendrites On the Surface of Iron-Doped Tin Oxide/Multiwalled Carbon Nanotube Nanocomposites

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Iron (Fe)-doped tin oxide/multiwalled carbon nanotube (SnO_2/MWCNT) nanocomposites was prepared by using the $SnCl_2$ solution method. SnO_2 nanowires, nanobelts, and nanodendrites were directly synthesized on the surface of Fe-doped SnO_2/MWCNT nanocomposites by calcination of the nanocomposites at 800 °C under an Ar atmosphere. The Fe-doped SnO_2/MWCNT nanocomposites and the prepared SnO_2 nanomaterials were characterized by field emission scanning electron microscopy, transmission electron microscopy, thermal gravimetric analysis, and X-ray diffraction. A growth mechanism of the SnO_2 nanomaterials was proposed and discussed. The as-prepared SnO_2 nanomaterials exhibited good gas-sensing properties and strong ultraviolet photoluminescence emissions. This technique provides a general, easy, and convenient approach for in situ preparation of SnO_2 nanomaterials on the desired substrates and nanodevices.

1. Introduction

One-dimensional (1-D) metal oxide nanostructures, such as nanotubes, nanowires, and nanobelts have attracted considerable attention due to their peculiar chemical and physical properties. Among these materials, tin oxide (SnO_2) is a very important material due to its potential application in many technological areas, such as electrocatalysts,[1] optical materials,[2] and electronic devices.[3] In particular, SnO_2 is a typical n-type wide band gap semiconductor (E_g = 3.6 eV at 300 K), and has been widely utilized as a gas-sensing material. Compared to the SnO_2-based film sensors, many researchers focused on the gas-sensing properties of SnO_2 nanostructured materials, such as nanoparticles,[4,5] single crystalline nanowires,[6–8] nanobelts,[9,10] nanorods,[5,11] polycrystalline nanotubes[12,13] or nanowire,[14] and hollow spheres.[15] Among the above SnO_2 nanomaterials, 1-D SnO_2 nanowires and nanobelts, with high surface-to-volume ratios, have attracted a great deal of attention, due to their potential applications in nanodevices. To date, many methods have been performed to synthesize SnO_2 nanowires and nanobelts, such as molten salt synthesis,[16,17] thermal oxidation of Sn nanowires,[18] thermal evaporation of metal or tin oxide powder,[19–22] and the carbothermal reduction method.[23–25] In general, thermal evaporation of metal or tin oxide powder needs quite high temperature (900−1380 °C)[19–21] or negative pressure (2−5 Torr).[22] The carbothermal reduction method can lower the reaction temperature to 700−850 °C.[23–25] Using the above two methods, SnO_2 nanomaterials were grown on the substrate which was placed downstream of the tube furnace. The growth of them can be explained by using a vapor−liquid−solid (VLS) or vapor−solid (VS) mechanism.

Since their discovery, carbon nanotubes (CNTs)[26] have attracted great interest as a new carbon material due to their unique chemical and physical properties.[27] Using CNTs as a reducing material, Marcelo Ornaghi Orlandi et al.[28] synthesized the SnO nanobelts and dendrites. In our previous work,[29] porous SnO_2 nanotubes with excellent gas-sensing properties were prepared by thermal treatment of the undoped SnO_2/MWCNT nanocomposites. In this work, iron (Fe)-doped SnO_2/MWCNT nanocomposites were synthesized by using the $SnCl_2$ solution method. The as-prepared wet nanocomposites were directly coated on Si, SiO_2, or ceramic substrates, and then calcinated at 800 °C under Ar atmosphere. Finally, SnO_2 nanowires, nanobelts, and nanodendrites were in situ synthesized on the surface of the Fe-doped SnO_2/MWCNT nanocomposites. The as-prepared SnO_2 nanomaterials exhibited good gas-sensing characteristics and excellent optical properties.

2. Experimental Section

2.1. Preparation of Fe-Doped SnO_2/MWCNT Nanocomposites and SnO_2 Nanomaterials. MWCNTs were purchased from Shenzhen Nanotech Port Co., Ltd. The diameter of the MWCNTs is 20−30 nm. The purification processes of the MWCNTs are the same as those found in our previous report.[29] In a typical synthesis of Fe-doped SnO_2/MWCNT nanocomposites, 1 g of tin(II) chloride ($SnCl_2 \cdot 2H_2O$) and 0.1 g iron nitrate($Fe(NO_3)_3 \cdot 9H_2O$) were dissolved in 40 mL of distilled H_2O, and then 0.25 mL of HCl (38%) was added. Subsequently, 10 mg of the MWCNTs was dispersed in the above solution. The mixture was sonicated for 20 min and then stirred at room temperature for 2 h. The Fe-doped SnO_2/MWCNT nanocomposites were then separated by centrifugation and washed with distilled water several times until the pH of the solution was neutral.[30,31]

For the growth of SnO_2 nanomaterials, small amounts of the as-prepared wet Fe-doped SnO_2/MWCNT nanocomposites were coated on Si, SiO_2, or ceramic substrates, dried in air, and then calcinated at 350 °C in air for 2 h. After that, the substrate was put in a quartz boat, inserted in the center of a quartz tube reactor, and then heated to 800 °C under Ar atmosphere at a

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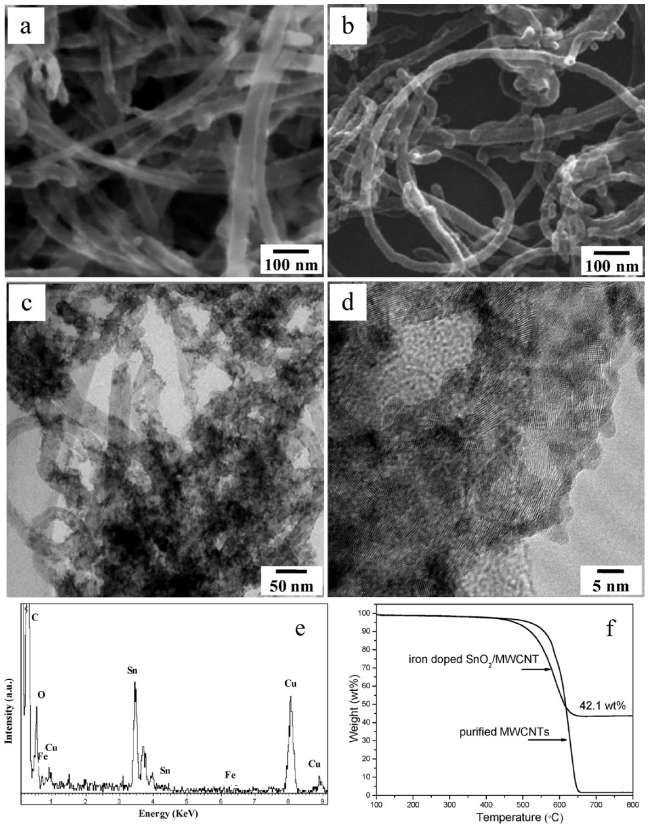

Figure 1. SEM images of the purified MWCNTs (a) and the prepared Fe-doped SnO_2/MWCNT nanocomposites (b). TEM (c), HRTEM (d) images, and the EDX spectrum (e) of the Fe-doped SnO_2/MWCNT nanocomposites. TGA curves (f) of the purified MWCNTs and the Fe-doped SnO_2/MWCNT nanocomposites.

flow rate of 200 sccm. After 2 h reaction, the system was allowed to cool down to room temperature in Ar atmosphere. Finally, SnO_2 nanomaterials were obtained on the substrate.

2.2. Characterization and Gas-Sensing Measurement. The Fe-doped SnO_2/MWCNT nanocomposites and the SnO_2 nanomaterials were characterized by field emission scanning electron microscopy (FE-SEM, Sirion 200, operated at 5 kV), X-ray diffraction (XRD, X'Pert Pro MPD, Cu Kα radiation, wavelength 1.5418 Å), transmission electron microscopy (TEM), and high-resolution transmission electron microscopy (HR-TEM, JEOL-2010, operated at 120 kV). The weight content of ferric and tin oxide in nanocomposites was measured by thermal gravimetric analysis (TGA, Pyris 1, heating rate 10 °C/min in flow air). Photoluminescence (PL) measurement was performed at room temperature using a He−Cd laser at a wavelength of 325 nm as excitation source. The structure of the sensor device and the measurement system are the same as our previous report.[29,32] The sensor response was defined as $S = R_a/R_g$, where R_a is the resistance in dry air and R_g is that in the dry air mixed with detected gases. In addition, the response time was defined as the time required for the conductance to reach 90% of the equilibrium value after a test gas was injected, and the recovery time was the time necessary for a sensor to attain a conductance 10% above its original value in air.

3. Results and Discussion

3.1. Structures of the Fe-Doped SnO_2/MWCNT Nanocomposites and SnO_2 Nanomaterials. Figure 1, parts a and b, shows SEM images of the purified MWCNTs and the Fe-doped SnO_2/MWCNT nanocomposites, respectively. The surface of the tube-like nanocomposites was obviously coarser than that of the purified MWCNTs. TEM and HRTEM images shown in Figure 1, parts c and d, suggest that the average crystallite size of SnO_2 nanoparticles was about 3−4 nm. The energy dispersive X-ray spectroscopy (EDX) spectrum shown in Figure 1e shows that the atomic content of Fe was 0.34%. Although the content of Fe is very low, it plays a very important role in the growth of SnO_2 nanomaterials. TGA curves shown in Figure 1f present the weight content of ferric and tin oxide in the nanocomposites was 42.1%.

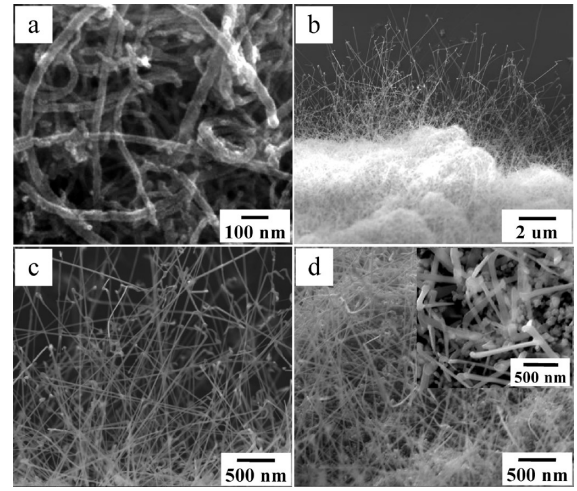

Figure 2. SEM image (a) of the Fe-doped SnO_2/MWCNT nanocomposites after calcination at 350 °C in air for 2 h; SEM images(b−d) of the SnO_2 nanowires and nanobelts after calcination the Fe-doped SnO_2/MWCNT nanocomposites at 800 °C under Ar atmosphere for 2 h.

Figure 2a is the SEM image of the nanocomposites after calcination at 350 °C in air for 2 h. The surface of the MWCNTs turned coarser than that of the as-prepared nanocomposites, which can be attributed to the growth of the SnO_2 nanoparticles. Subsequently, the nanocomposites were calcined at 800 °C under Ar atmosphere for 2 h, and the SEM images of the products are shown in Figure 2b−d. It is clear that large amounts of nanowires and nanobelts were obtained on the surface of the nanocomposites. The average diameter of the nanowires was about 50 nm. The particle on the tip of the nanowire indicates the growth of them can be considered to be VLS mechanism. XRD patterns shown in Figure 3 suggest the obtained nanowires and nanobelts were rutile

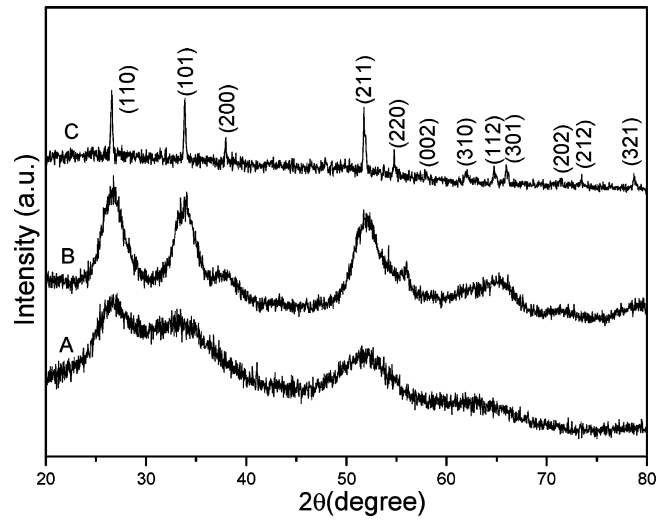

Figure 3. XRD patterns of the Fe-doped SnO_2/MWCNT nanocomposites before (A) and after (B) calcination at 350 °C in air for 2 h, and the SnO_2 nanowires (C).

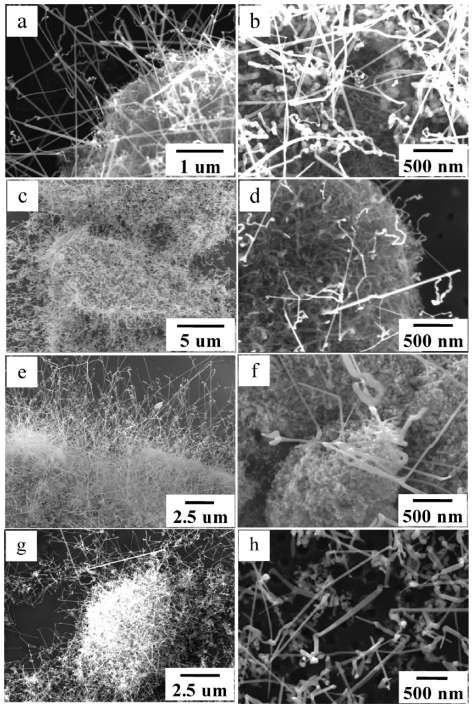

Figure 4. SEM images of the products after calcination the Fe-doped SnO_2/MWCNT nanocomposites for 15 min (a,b), 30 min (c,d), 1 h (e,f), and 3 h (g,h).

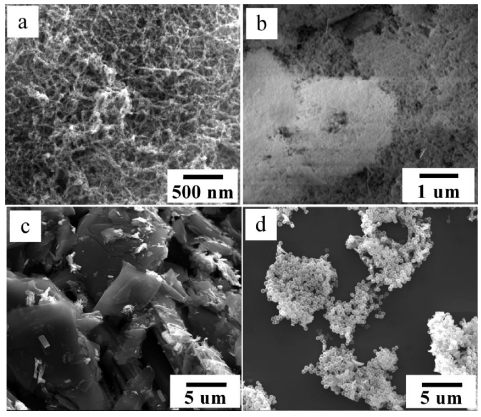

Figure 5. SEM image of the products grown from the undoped SnO_2/MWCNT nanocomposites (a), and SnO_2/MWCNT nanocomposites after immerged in 0.1 mol/L of iron nitrate solution for 1 min (b). SEM images of the products after calcination the Fe-doped SnO_2/graphite powder (c) and SnO_2/carbon sphere (d) nanocomposites.

SnO_2 (JCPDS 77−0448). It should be mentioned that the SnO_2 nanowires and nanobelts were grown in situ on the surface of Fe-doped SnO_2/MWCNT nanocomposites during the thermal treatment, and at the same time, the MWCNTs disappeared. Figure 3 also show the XRD patterns of the Fe-doped SnO_2/MWCNT nanocomposites before and after calcination at 350 °C in air for 2 h. The diffraction peaks of SnO_2 turned sharper after calcination, which can be attributed to the improved crystallization and the growth of the SnO_2 nanoparticles.

3.2. Growth Mechanism of the SnO_2 Nanomaterials. In order to clarify the growth process of the SnO_2 nanowires and nanobelts, the products grown at the initial stage were investigated by SEM, and the results are shown in Figure 4. After calcination the Fe-doped SnO_2/MWCNT nanocomposites for 30 min, a large amount of SnO_2 nanowires and nanobelts were observed. Figure 4, parts b and d, confirms the SnO_2 nanowires and nanobelts were grown in situ on the surface of the nanocomposites. After 1 h calcination, all of the MWCNTs were used up and many nanoparticles were observed at the same time, as shown in Figure 4e,f. The results suggest MWCNTs were oxidized by SnO_2 at high temperature, and SnO_2 was synchronously reduced. The reduced product agglomerated and formed large nanoparticles. Further prolonging the calcination time, most of the nanoparticles were transformed to SnO_2 nanowires and nanobelts, as shown in Figure 2b−d and Figure 4g,h. The results mean the nanoparticles shown in Figure 4e,f were unstable, and further imply that these nanoparticles were SnO which result from the reaction of the MWCNTs and SnO_2.

Figure 5a presents the SEM image of the products after calcination the undoped SnO_2/MWCNT nanocomposites. No SnO_2 nanowires or nanobelts were obtained. The result suggests that a small amount of Fe plays a very important role in the growth of 1-D SnO_2 nanomaterials. In order to confirm the role of Fe, the undoped SnO_2/MWCNT nanocomposites were immerged in 0.1 mol/L of iron nitrate solution for 1 min, and then dried in air. After the same thermal treatment process, the corresponding products are shown in Figure 5b. Though only small amount of SnO_2 nanowires were obtained, it ascertains the important role of Fe in the growth of SnO_2 nanowires or nanobelts. In addition, the MWCNTs in the nanocomposites were replaced with graphite powder and carbon spheres,[32] and the SEM images of the corresponding products are shown in Figure 5c,d. No SnO_2 nanowires and nanobelts were obtained, which may be result from the different microstructures of the carbon materials.

In order to investigate the growth mechanism of the 1-D SnO_2 nanomaterials, the intermediate products shown in Figure 4c,d were further characterized by TEM, and the results are shown in Figure 6. In Figure 6a, it is clear that some nanoparticles were agglomerated together and formed large particles. The EDX result suggests that the atomic ratio of Sn to O in these particles was about 47/52, which was very close to SnO. The atomic content of Fe was 0.64%, larger than that of Fe-doped SnO_2/MWCNT nanocomposites. The results mean that SnO_2 was reduced to SnO by MWCNTs. At the same time, ferric oxide was also reduced to Fe. It is well-known that SnO is

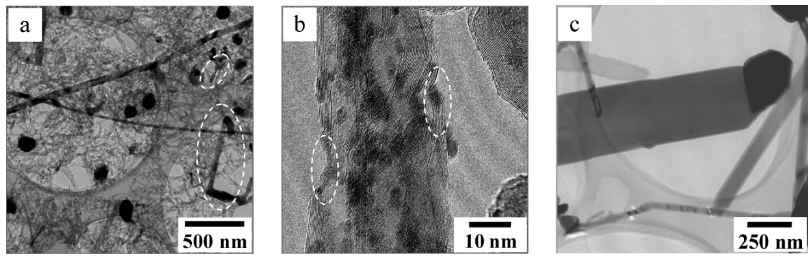

Figure 6. TEM image (a) of the products after calcination the Fe-doped SnO_2/MWCNT nanocomposites for 30 min. HRTEM image (b) of the etched MWCNTs. TEM image (c) of the SnO_2 nanobelt with a catalyst particle at the tip.

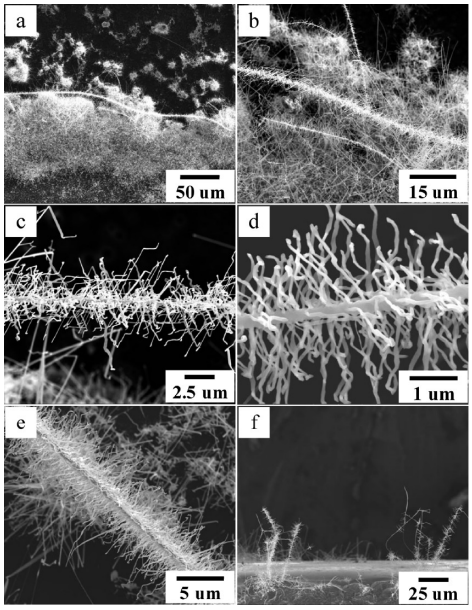

Figure 7. SEM images of the SnO_2 nanodendrites grown with 4 h calcination time.

metastable and will decompose into SnO_2 and Sn. The Sn and Fe formed Sn−Fe alloyed droplets, and these alloyed droplets can provide the energetically favored sites for adsorption of SnO vapor. Subsequently, the decomposition of SnO resulted in the precipitation of SnO_2, and as a result, the SnO_2 nanowires and nanobelts were formed, as shown in Figure 6a. So, the SnO_2 nanowires and nanobelts were in situ synthesized on the surface of the nanocomposites via a VLS mechanism.[23] Figure 6b presents the HRTEM image of the MWCNTs. The sidewall of MWCNTs was clearly etched by the metal oxide nanoparticles. Figure 6c is the TEM image of the SnO_2 nanobelts with a catalyst particle at the tip. The EDX result suggests, in the catalyst particle, that the atomic content of Fe was 2.67%, which is 8 times greater than that of the buck nanocomposites. In addition, the atomic ratio of Sn to O in it was about 65/33. The results suggest the existence of Sn, and also mean the formation of Sn−Fe alloy.

Furthermore, when the calcination time was prolonged to 4 h, three-dimensional (3-D) SnO_2 nanodendrites, several hundred micrometers in length, were obtained, as shown in Figure 7. The formation of the SnO_2 nanodendrites implies the secondary and the tertiary growth of the SnO_2 nanowires on the primary SnO_2 nanostructures. When compared to the 1-D nanostructures shown in Figure 2, there were no particles at the tip of the branches. The results suggest the VS mechanism for the secondary and the tertiary growth of nanowires, which was similar to the multistep thermal vapor deposition process.[33,34]

3.3. Gas-Sensing and Optical Properties. The as-prepared wet Fe-doped SnO_2/MWCNT nanocomposites were directly coated on the outer surface of the ceramic tube and dried in air. After thermal treatment, SnO_2 nanowires and nanobelts were grown in situ on the surface of the ceramic tube. The gas-sensing properties were investigated, and the results are shown in Figure 8. Figure 8a,b shows the real-time response curve and the sensor responses of the sensor device upon exposure to different concentrations of ethanol at a working temperature of 200 °C, respectively. The sensor fabricated with the SnO_2 nanowires and nanobelts show a good response and reversibility to ethanol. The sensing response to 100 ppm of ethanol was 24, which

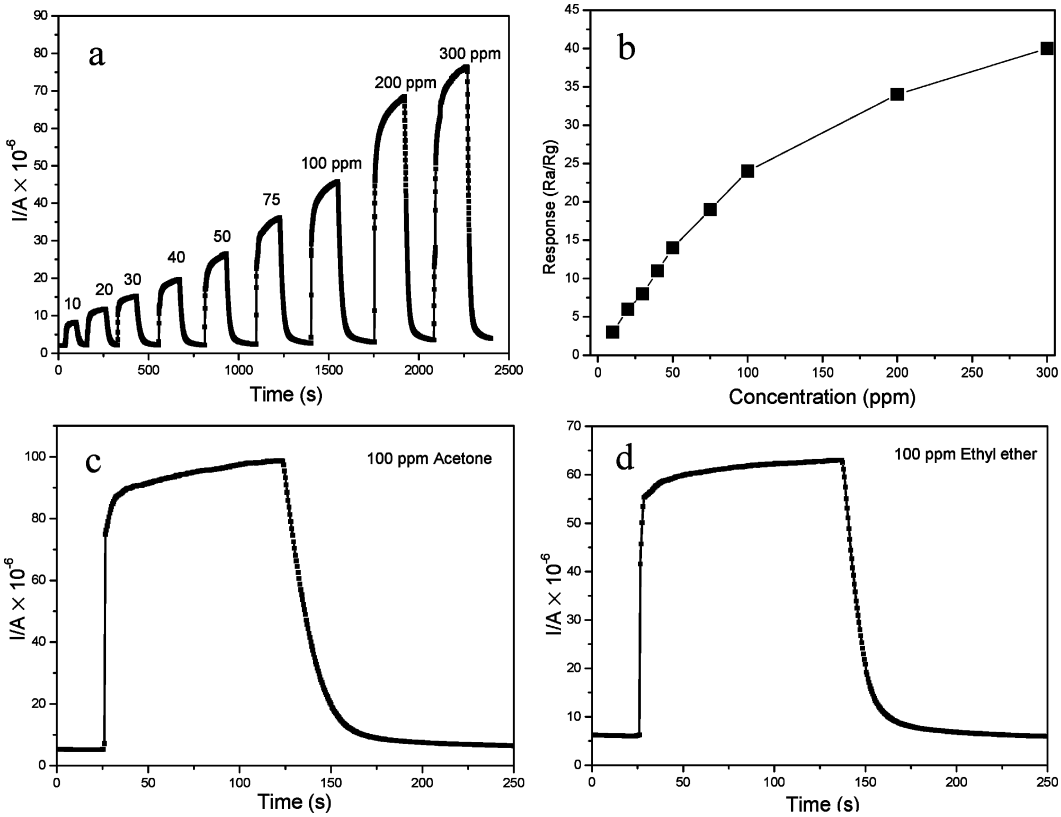

Figure 8. Gas-sensing properties of the as-prepared SnO_2 nanowires and nanobelts. Real-time response curve (a) and the sensor responses (b) of the sensor device upon exposure to different concentrations of ethanol at a working temperature of 200 °C. Real-time response curve of the sensor device upon exposure to 100 ppm of acetone (c) and ethyl ether (d) at a working temperature of 200 °C.

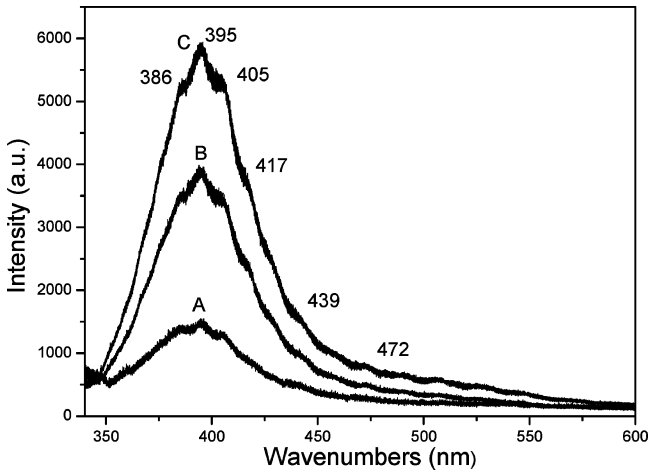

Figure 9. Room-temperature PL spectra of SnO_2 nanostructures prepared with 1(A), 2(B), and 4 h (C) calcination times.

was higher than those of SnO_2 nanobelts,[10] SnO_2 nanorods,[11] Sb-doped 1-D,[35] and branched[36] SnO_2 nanowires, and was lower than that of La_2O_3 doped SnO_2 nanomaterials.[37,38] The response and recovery time were about 10 and 15 s, respectively. When increasing the concentration, the response of the sensor also linearly increased in the range of low concentration, as shown in Figure 8b. In addition, the sensor also exhibits a good response to acetone and ethyl ether, as shown in Figure 8c,d. The sensing responses to 100 ppm of acetone and ethyl ether were 20 and 11, respectively. The response times were less than 3 s.

The PL spectra of the obtained SnO_2 nanostructures grown with different calcination times were measured at room temperature, and the results are shown in Figure 9. There are four strong emission bands at 386, 395, 405, and 417 nm, and two weak peaks at 439 and 472 nm. Previous reports indicated that SnO_2 thin films exhibit a broad dominant peak near 396 nm.[39] The peaks at 405 and 439 nm are due to structural defects or luminescent centers, such as nanocrystals and defects in the SnO_2 nanowires and nanobelts.[40,41] The 472 nm peak is possibly attributed to the electron transition mediated by defect levels such as oxygen vacancies in the band gap, and the peak at 417 nm may be caused by other defects or oxygen vacancies.[42] The 386 nm peak is close to the ultraviolet peak of the SnO_2 nanowires reported by Ayan Kar et al., which was due to the mergence of the band to acceptor and donor to acceptor peaks.[43] Furthermore, with the increasing of the calcination time, the intensities of PL emissions were also increased. In Figure 9, lines A, B, and C are the PL spectra of the SnO_2 nanomaterials prepared with 1, 2, and 4 h calcination time, respectively. All of the as-prepared SnO_2 nanomaterials have similar PL emission peaks. On the basis of the results of the SEM images, the SnO_2 nanodendrites have the strongest PL emission intensities, which may result from the increased structural defects or luminescent centers between the trunk and the branches.[44]

4. Conclusions

SnO_2 nanowires and nanobelts were in situ synthesized on the surface of Fe-doped SnO_2/MWCNT nanocomposites by calcination of the nanocomposites at 800 °C under an Ar atmosphere. The growth of the SnO_2 nanowires and nanobelts was discussed on the basis of the VLS mechanism. 3-D SnO_2 nanodendrites were also synthesized by a multistep VS procedure on the primary nanomaterials. The as-prepared SnO_2 nanomaterials exhibit good gas-sensing properties and strong ultraviolet emissions. The results provide a new route for in situ synthesis of 1-D and 3-D SnO_2 nanomaterials with attractive gas-sensing and optical properties on some desired substrates and nanodevices.

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